恩智浦半導體

公司簡介

2026

NXP

Brighter Together

共生、共好、共創

企業概述

以強大的優勢服務全球26,000多家客戶

恩智浦半導體(NXP Semiconductors N.V.)是一家總部位於荷蘭恩荷芬(Eindhoven)的上市公司，在全球各地都有營運據點。

恩智浦擁有超過33,100多名員工，敬業奉獻、齊心協力，懷著建構解決方案而非僅僅產品的熱情，致力於提升人類、團體與整個社會的能力。

超過60年

經驗和專業知識

超過9,500個

專利系列

33,100多名

優秀人才

11,600多名

研發人員

業務遍及

30多個國家/地區

126.1億美元

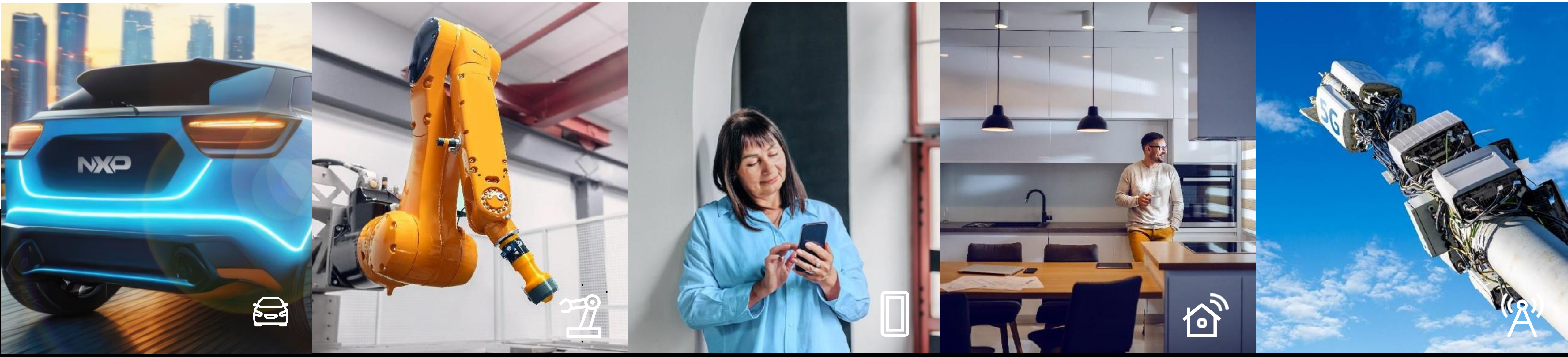
年營收[1]

[1] 2024年營收——請參考恩智浦網站 www.nxp.com/investor 瞭解更多資訊。

我們攜手並進 開發 突破性 解決方案

我們預先考慮「未來」需求，匯聚科技領域精英才智，創造具顛覆性的解決方案，推動產業持續向前，因應快速變化的世界

恩智浦半導體（NASDAQ：NXPI）身為值得信賴的合作夥伴，持續針對汽車、工業與物聯網、行動裝置與通訊基礎設施市場，提供創新解決方案。秉持「共生、共好、共創（Brighter Together）」企業理念，恩智浦致力創造領先業界的尖端技術並匯聚精英才智，開發系統解決方案，讓互聯世界變得更美好、更安全、更有保障。恩智浦在全球逾30個國家設有業務機構，2024年公司全年營業額達到126.1億美元。更多恩智浦相關訊息，請參閱官方網站：www.nxp.com。



Brighter Together

nxp.com

賽題分享

整合AI與創新邊緣運算技術
發揮智慧生活(Brighter Lives)的無限可能

- 參賽隊伍需使用恩智浦與安富利提供的MCX -N FRDM進行創作
- 主軸:智慧生活應用
- 恩智浦產品應用完整度(50分, GUI)
- 創意性與主題呈現度(智慧應用、安全連結、永續發展)(50分)
- 加分項: 使用Wi-Fi模組
- 第一名: 45,000,
- 第二名: 25,000,
- 第三名: 15,000

面向萬物互聯的下一哩路：AI 與邊緣運算在
跨域整合的無限可能

- 參賽隊伍需使用恩智浦與安富利提供的FRDM-IMX93進行創作
- 主軸:跨域整合應用
- 技術與系統整合能力（60 分)

  AI整合與最佳化能力、系統完整性、
人機體驗感受、GoPoint
- 創意性與主題呈現度（40 分）
- 加分項: Security
- 第一名: 45,000,
- 第二名: 25,000,
- 第三名: 15,000

Make NTU
FRDM – IMX93

FAE / Johnson Wu

Johnson.hx.wu@nxp.com

Offering platform

FRDM i.MX93 Board

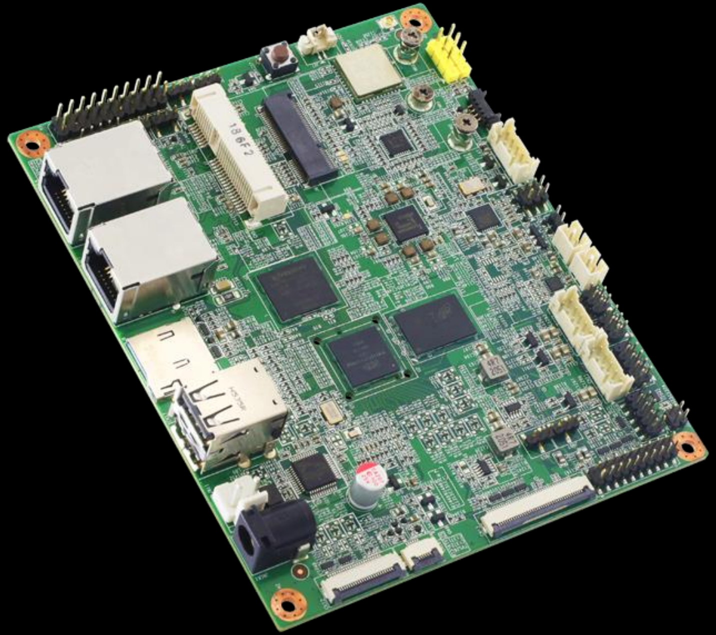
i.MX Development Platform Offering

From NXP

Evaluation Kits ranging from $300 to $1200+

From Partners

Evaluation Kit from $100 - $500+

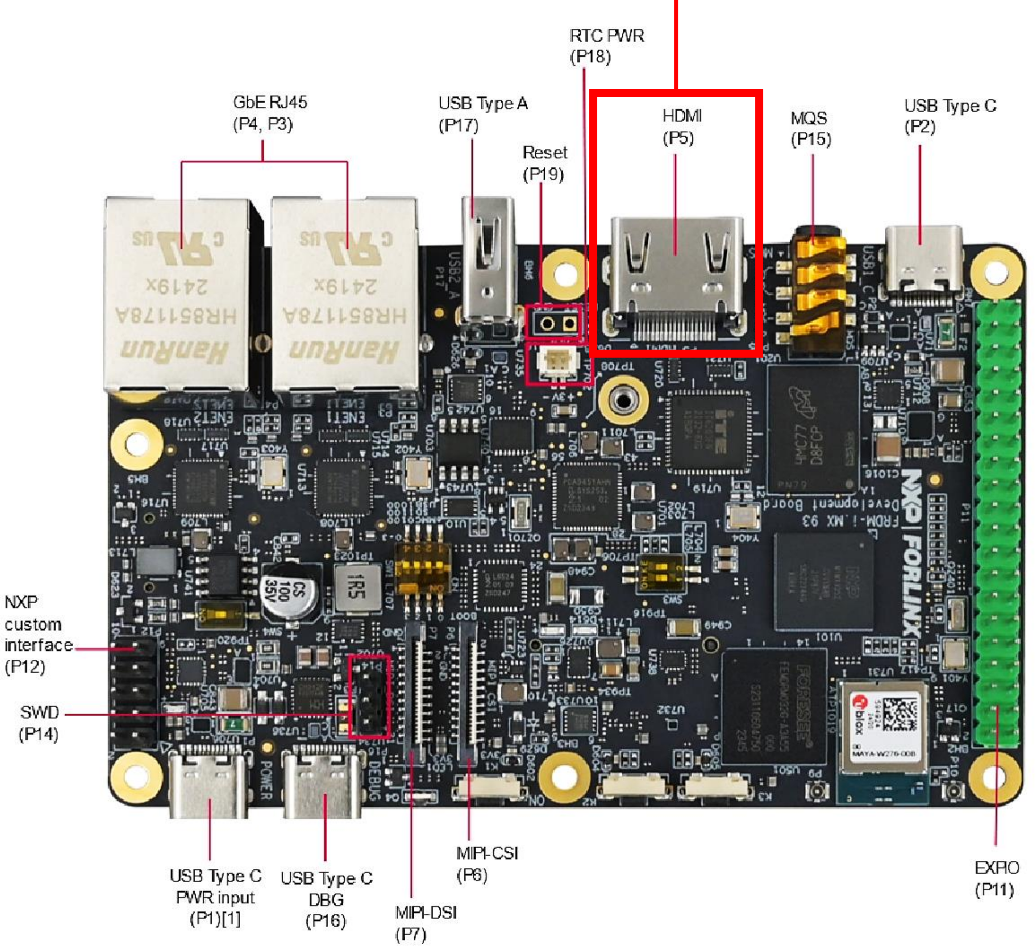


NXP Mass Market need

FRDM i.MX from $50~200

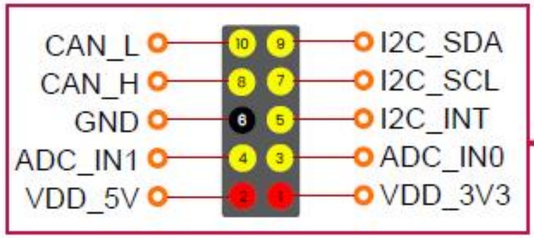
FRDM i.MX93 Board – Front

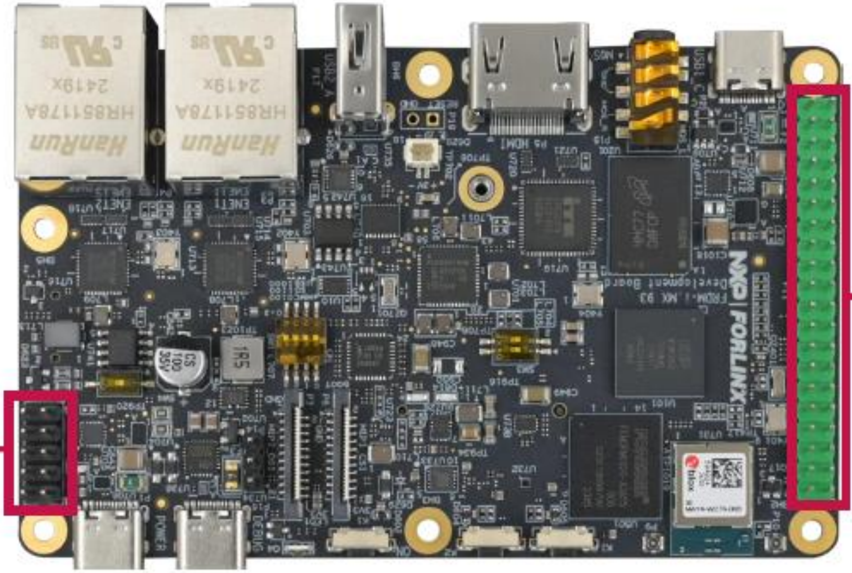
提醒：建議自行攜帶**HDMI**螢幕

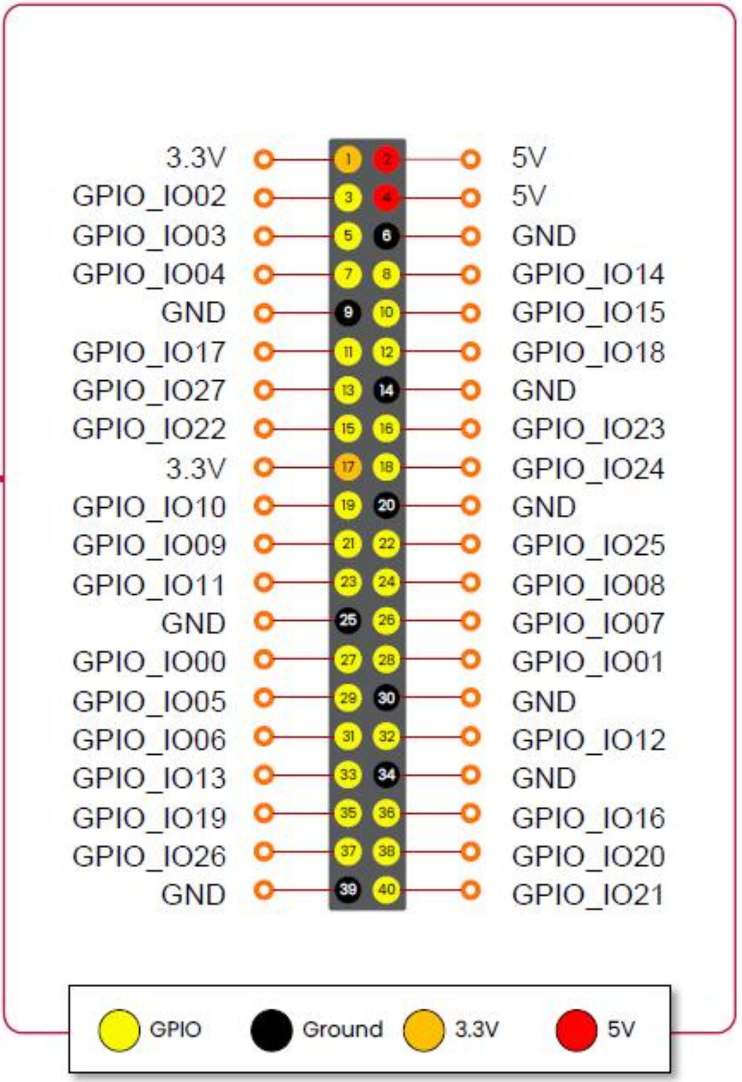


[1] - USB Type C PWR input (P1) shown in the figure is the only power supply port, and must always be supplied for system runnir

FRDM i.MX93 Board – Front

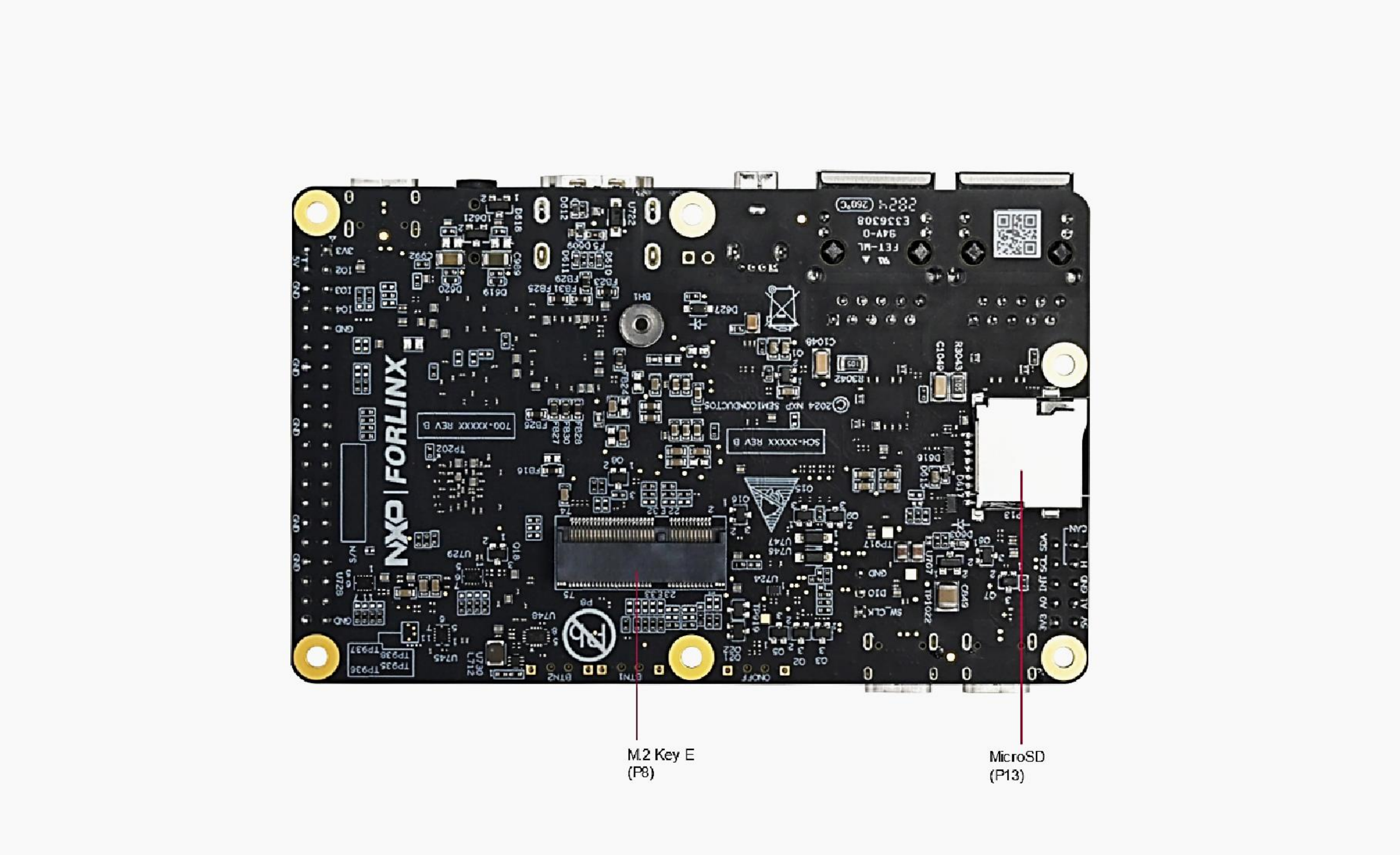






2x20 Pin connector can easily reuse peripherals on the market.

FRDM i.MX93 Board – Back



Make NTU

評分項 - Must

- 技術與系統整合能力（60 分)
  - AI整合與最佳化能力(0.5 TOP)、系統完整性、人機體驗感受、GoPoint(example)

- 創意性與主題呈現度（40 分）

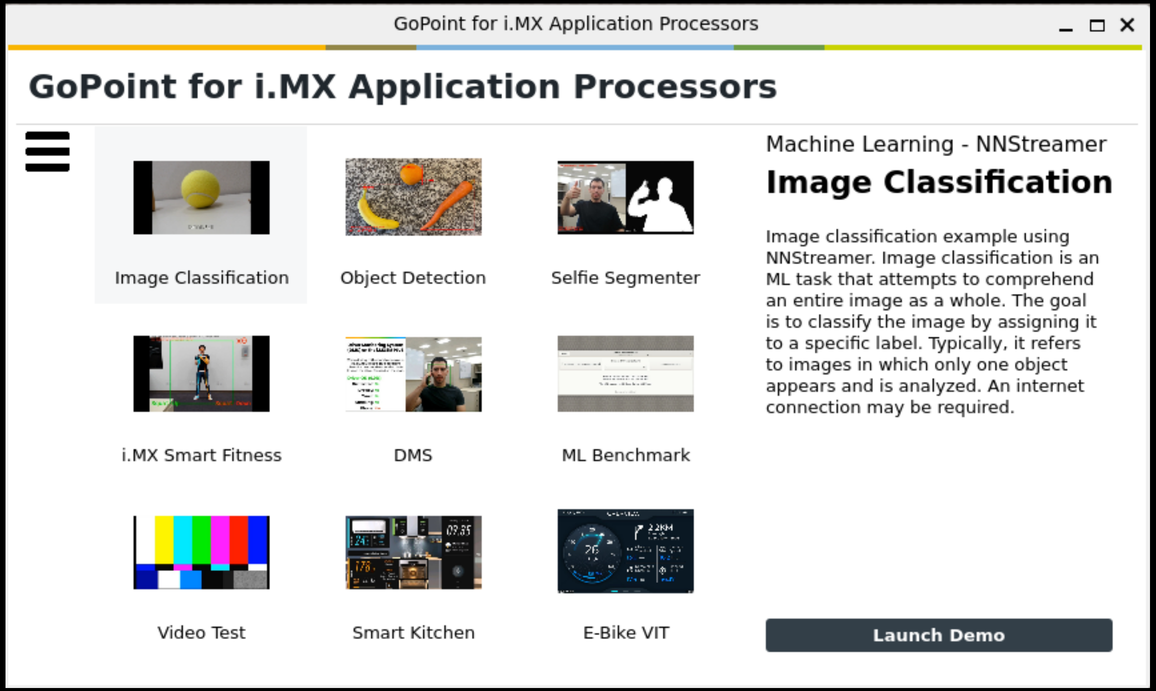
加分項

- Security
  - 不限形式的加密
  - 可在前端、平台端、資訊傳送時，做有意義的加密

GoPoint Demo On i.MX93 FRDM Board

GoPoint

- GoPoint is a user-friendly application that allows the user to launch preselected demonstrations included in the NXP provided BSP
- GoPoint follows the quarterly release roadmap for BSP
- How to launch GoPoint



GoPoint Demo On i.MX93 FRDM Board

- Since i.MX93 FRDM board's BSP is based on standard BSP release, GoPoint is included in i.MX93 FRDM Yocto build by default.
- List of 9 demos available on i.MX93 FRDM Board

- 1. Image Classification
- 2. Object Detection
- 3. Selfie Segmenter
- 4. i.MX Smart Fitness
- 5. DMS (Driver Monitor System)
- 6. ML Benchmark
- 7. Video Test
- 8. i.MX Smart Kitchen
- 9. i.MX E-Bike VIT

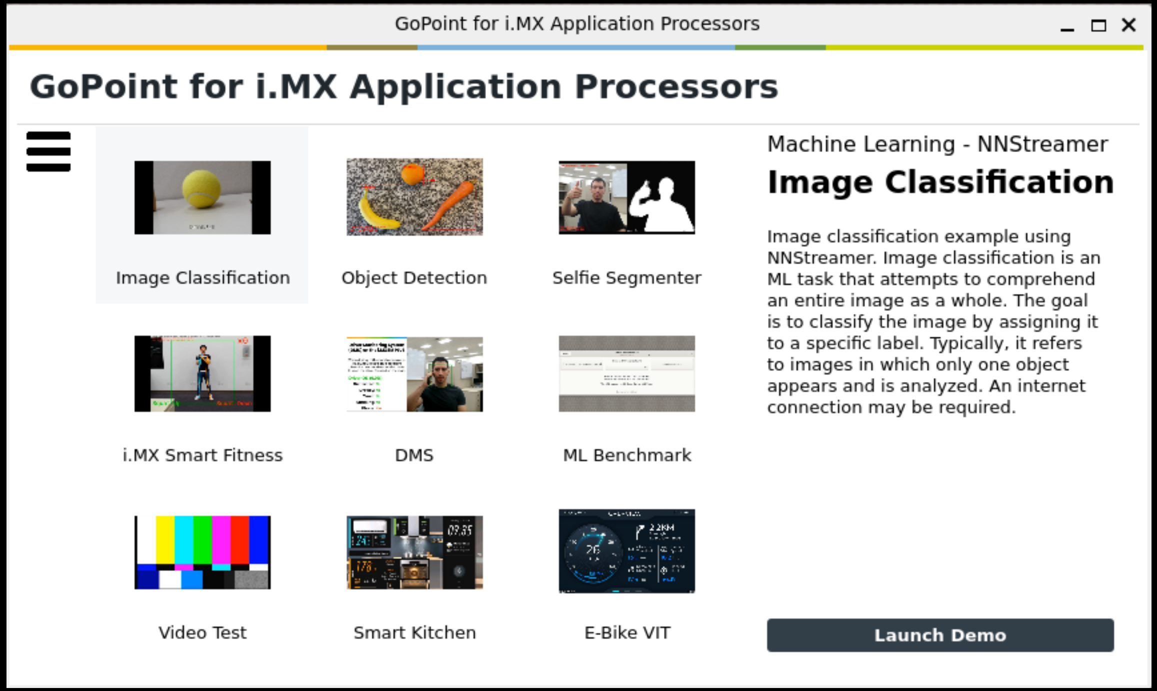
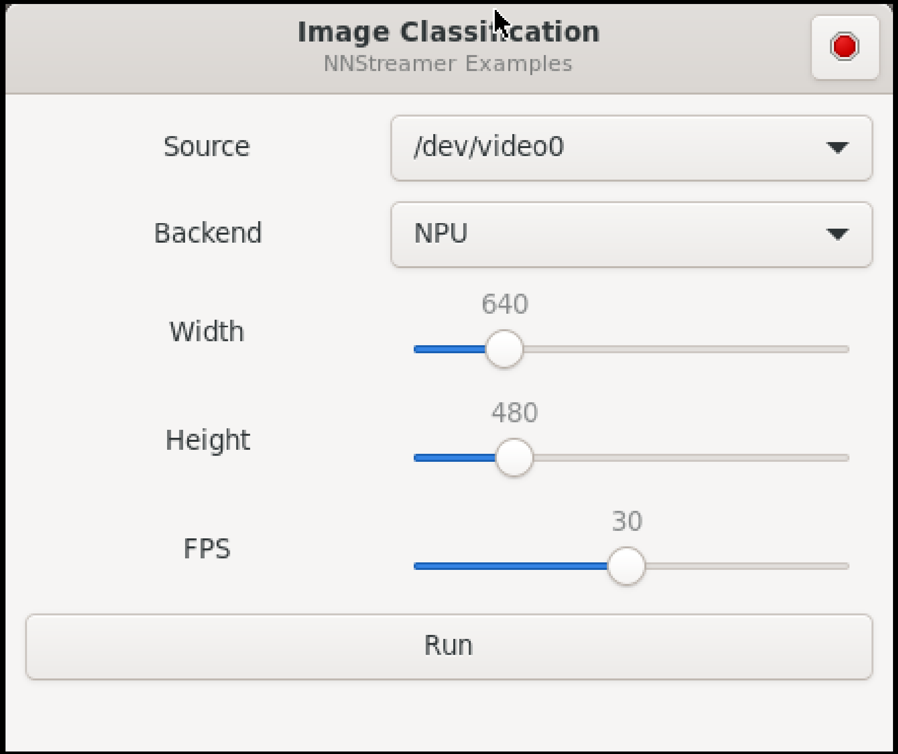


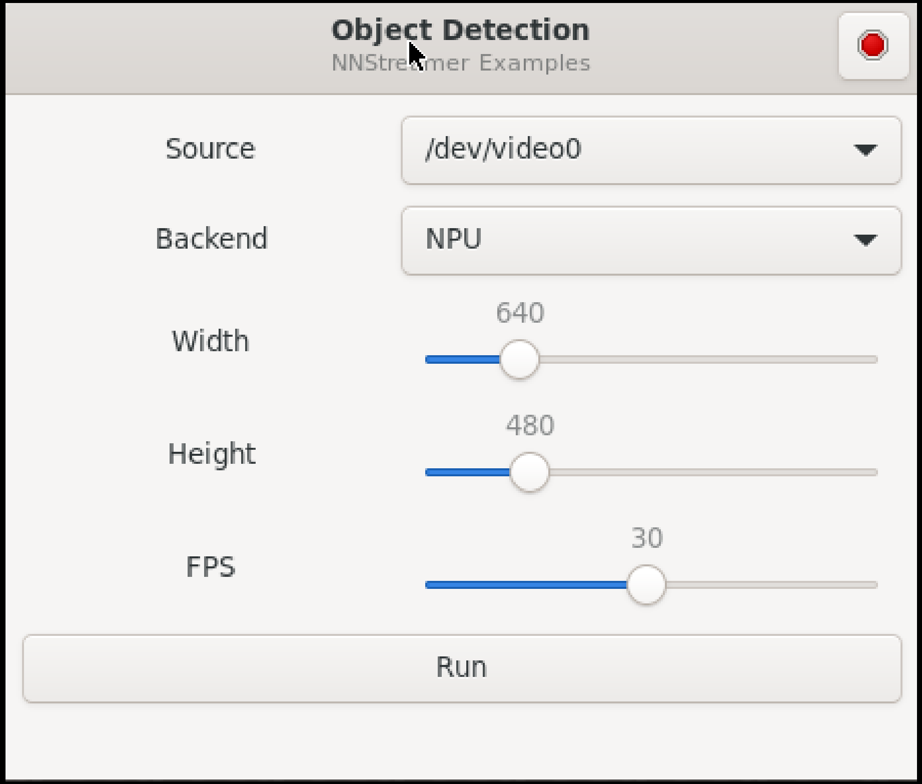
Image Classification Demo

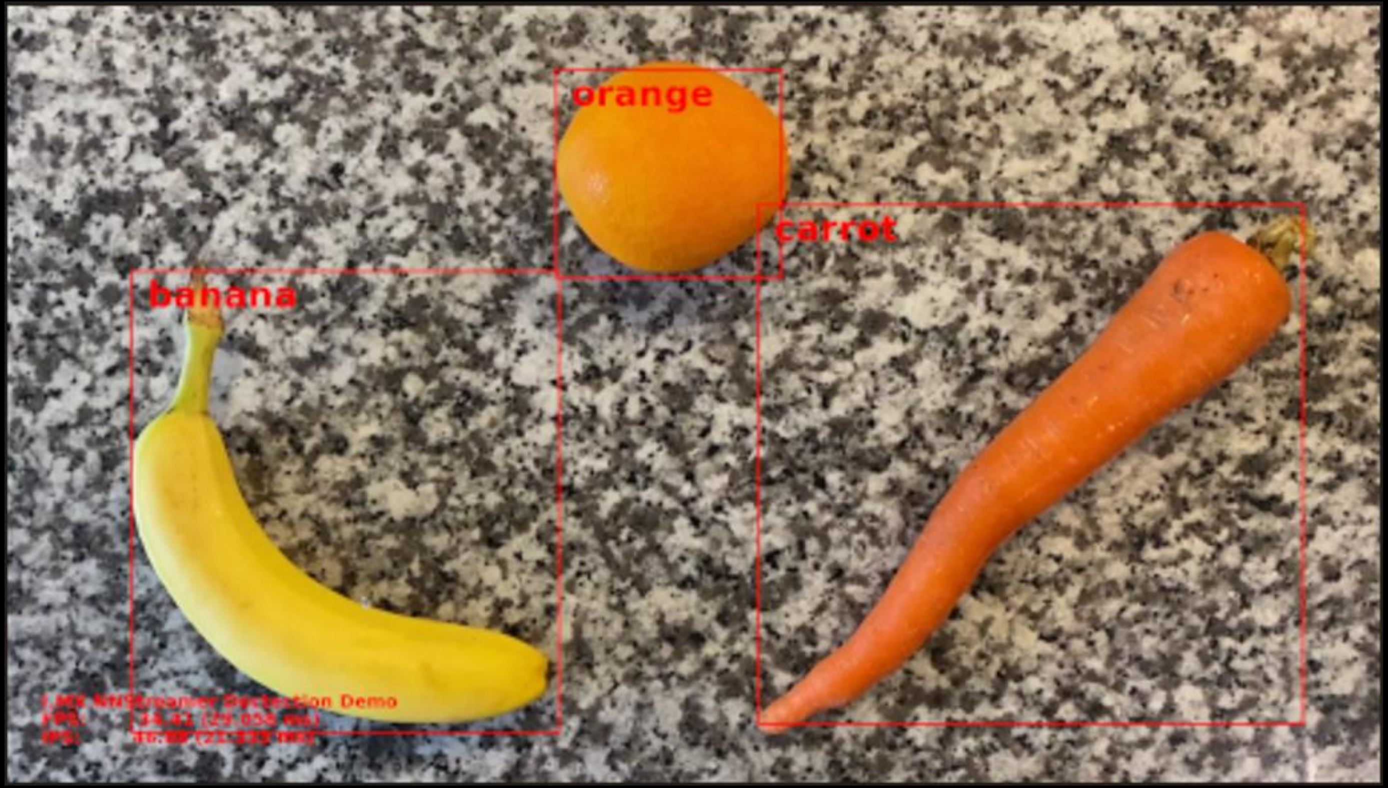
- Image classification is a ML task that attempts to comprehend an entire image as a whole. The goal is to classify the image by assigning it to a specific label. Typically, it refers to images in which only one object appears and is analyzed. This example is using NNStreamer.



Object Detection Demo

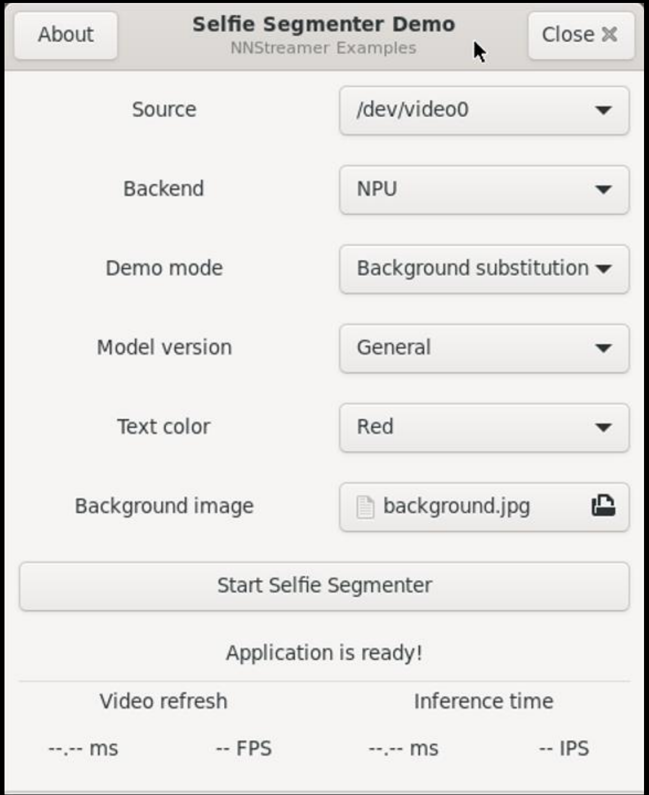
- Object detection is the ML task that detects instances of objects of a certain class within an image. A bounding box and a class label are found for each detected object. This example is using NNStreamer.





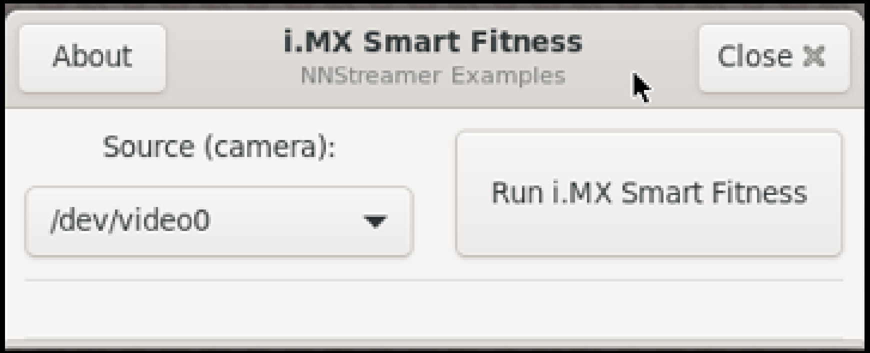
Selfie Segmenter Demo

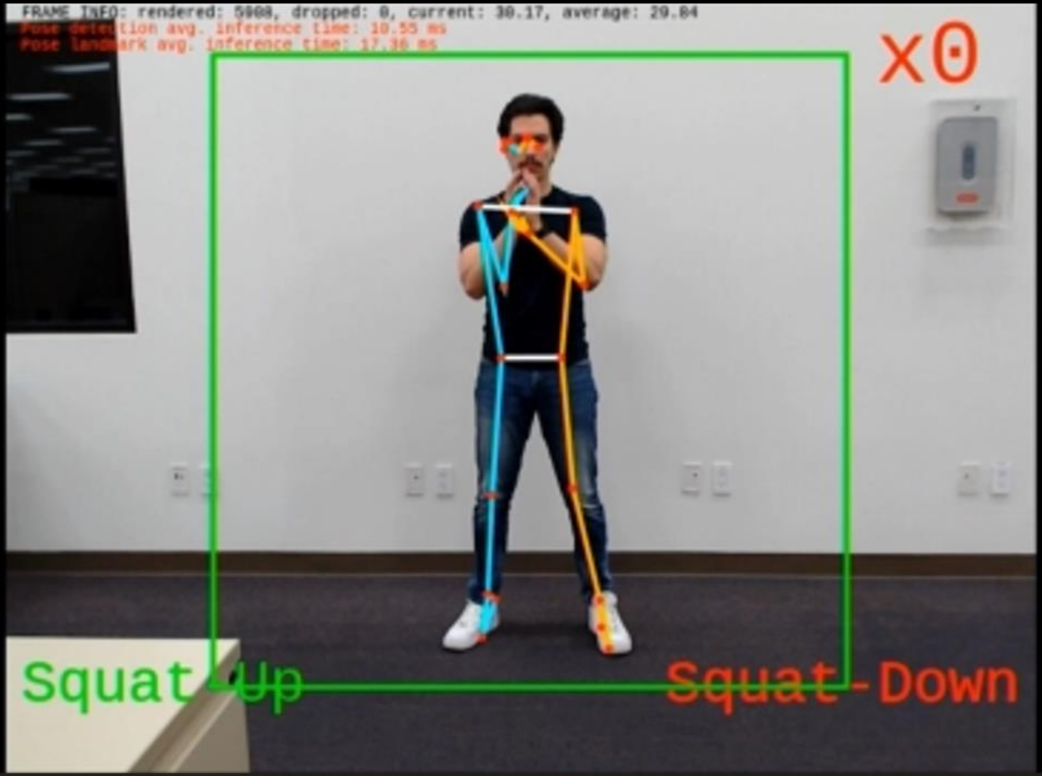
- Selfie Segmenter showcases the ML capabilities of i.MX 93 by using the NPU to accelerate an instance segmentation model. This model lets you segment the portrait of a person and can be used to replace or modify the background of an image. This example is using NNStreamer.



i.MX Smart Fitness Demo

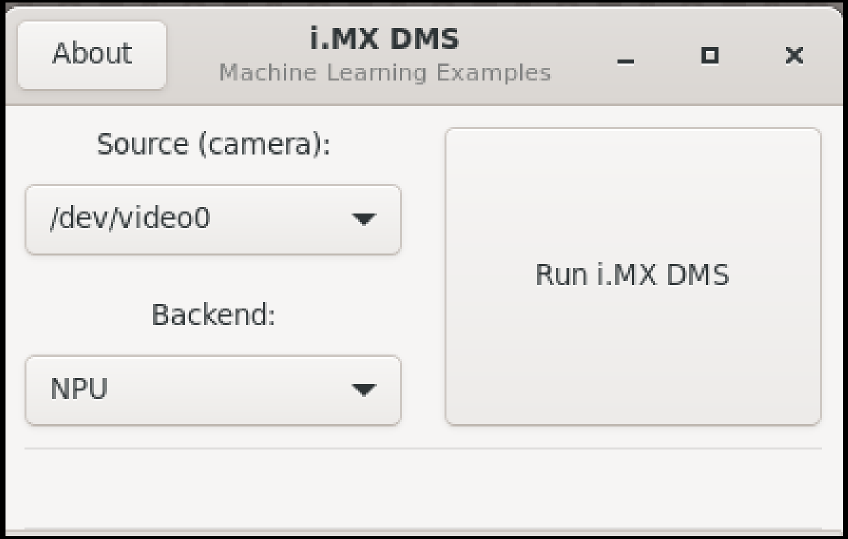
- i.MX Smart Fitness showcases the i.MX' Machine Learning capabilities by using an NPU to accelerate two Deep Learning vision-based models. Together, these models detect a person present in the scene and predict 33 3D-keypoints to generate a complete body landmark, known as pose estimation. From the pose estimation, the application tracks the 'squats' fitness exercise.

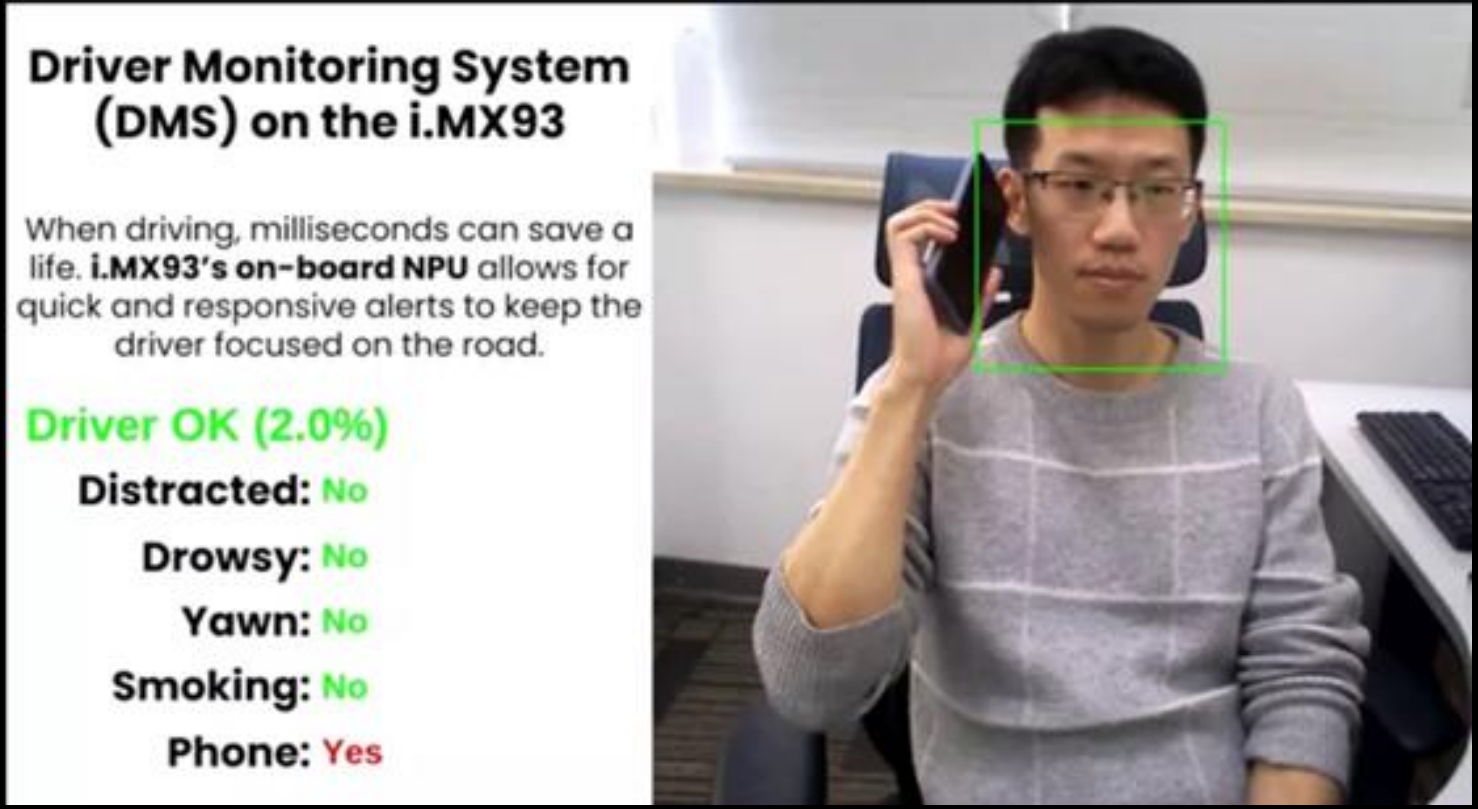




DMS (Driver Monitor System) Demo

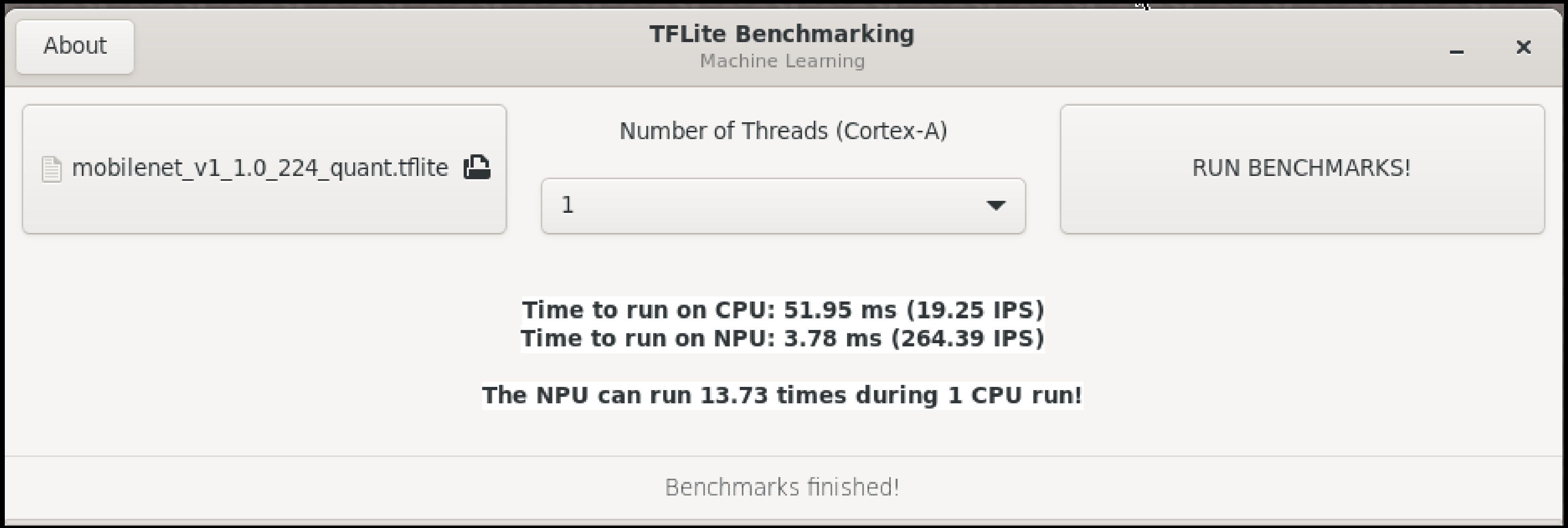
- This application showcases the capability of implementing DMS on i.MX 93 platform, and the performance boost brought by Neural Processing Unit (NPU). DMS uses four ML models in total to achieve face detection, capturing face landmark and iris landmark, smoking detection and calling detection.





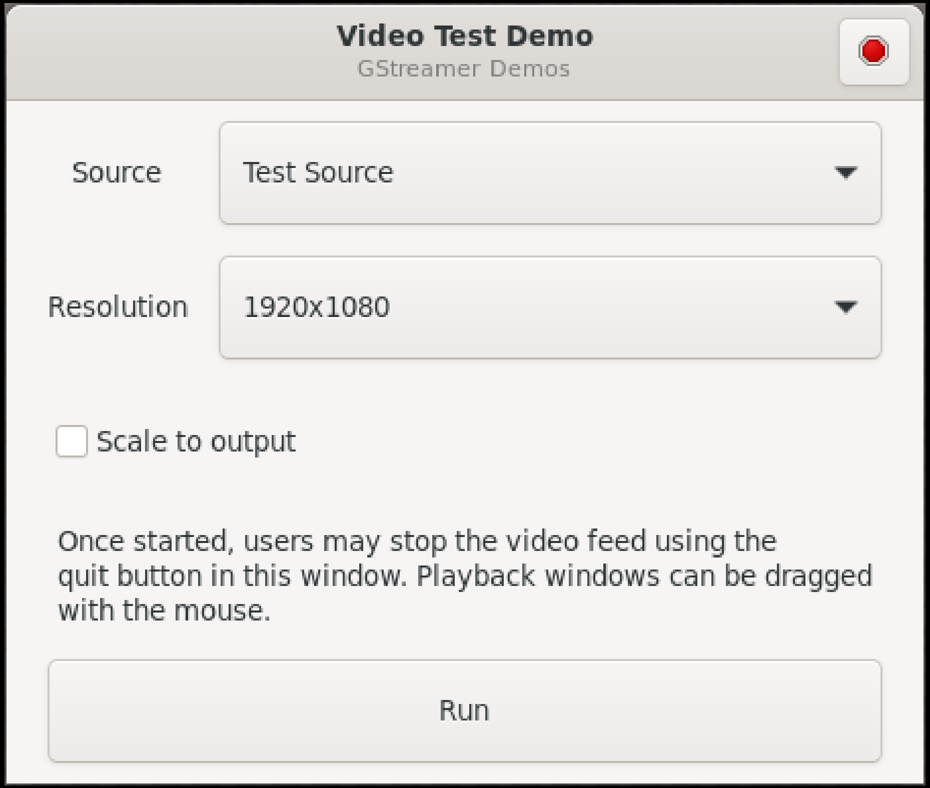
ML Benchmark Demo

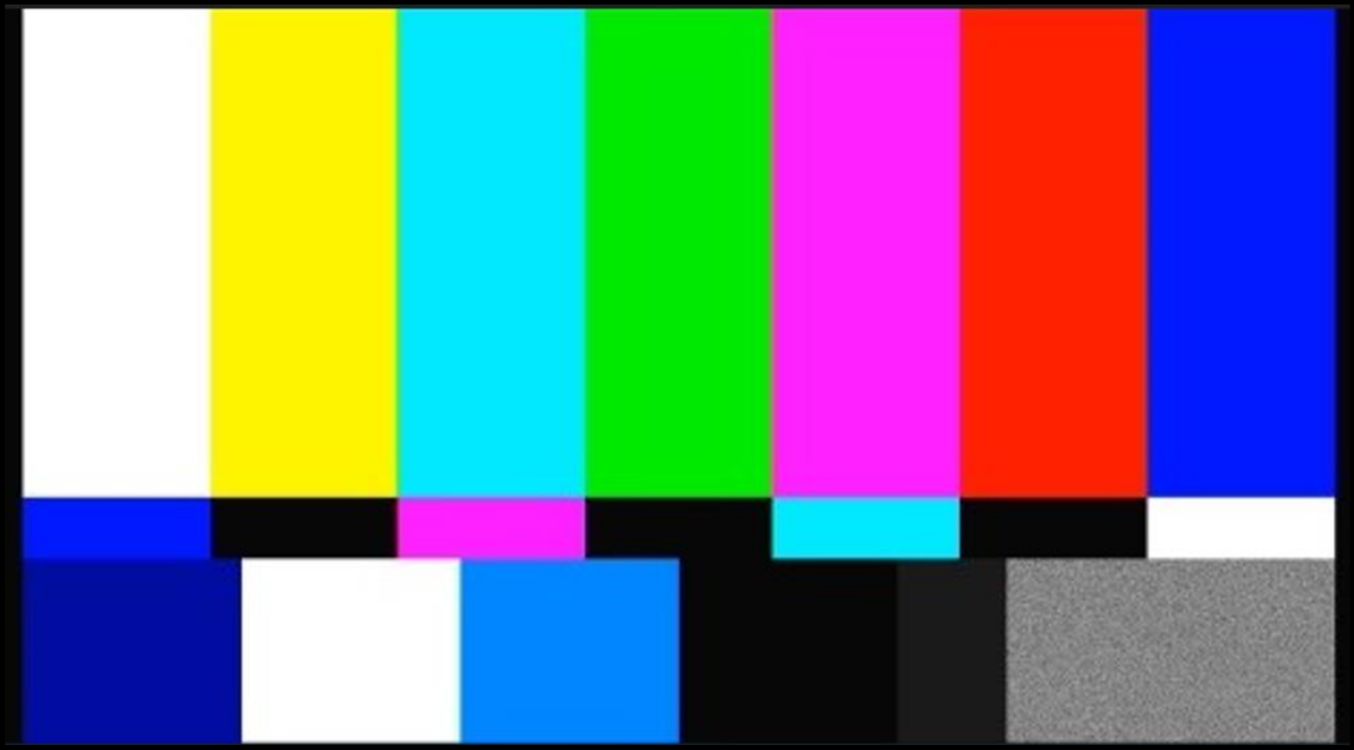
- This example is based on benchmark_model tool in Tensorflow Lite framework, which allows to easily compare the performance of TensorFlow Lite models running on CPU (Cortex-A) and NPU.



Video Test Demo

- This is a simple demo that allows users to play back video captured on a camera or a test source. It’s based on gstreamer pipeline.





i.MX Smart Kitchen Demo

- i.MX Smart Kitchen showcases the Multimedia capabilities of i.MX to emulate an interactive kitchen through a GUI controlled by voice commands. The GUI is based on LVGL (Little Versatile Graphic Library) and NXP's Voice Intelligent Technology (VIT) supports the voice commands.
- Usage: Keyword + command

Wake words supported
HEY HOOD
HEY OVEN
HEY AIRCON

General commands
(work with any wake word)

ENTER
EXIT
RUN DEMO
STOP DEMO

Hood commands

FAN OFF
FAN ON
FAN LOW
FAN HIGH
LIGHT OFF
LIGHT ON

Aircon commands

FAN LOW
FAN HIGH
DRY MODE
COOL MODE
FAN MODE
SWING OFF
SWING ON

Oven commands

CLOSE DOOR
OPEN DOOR

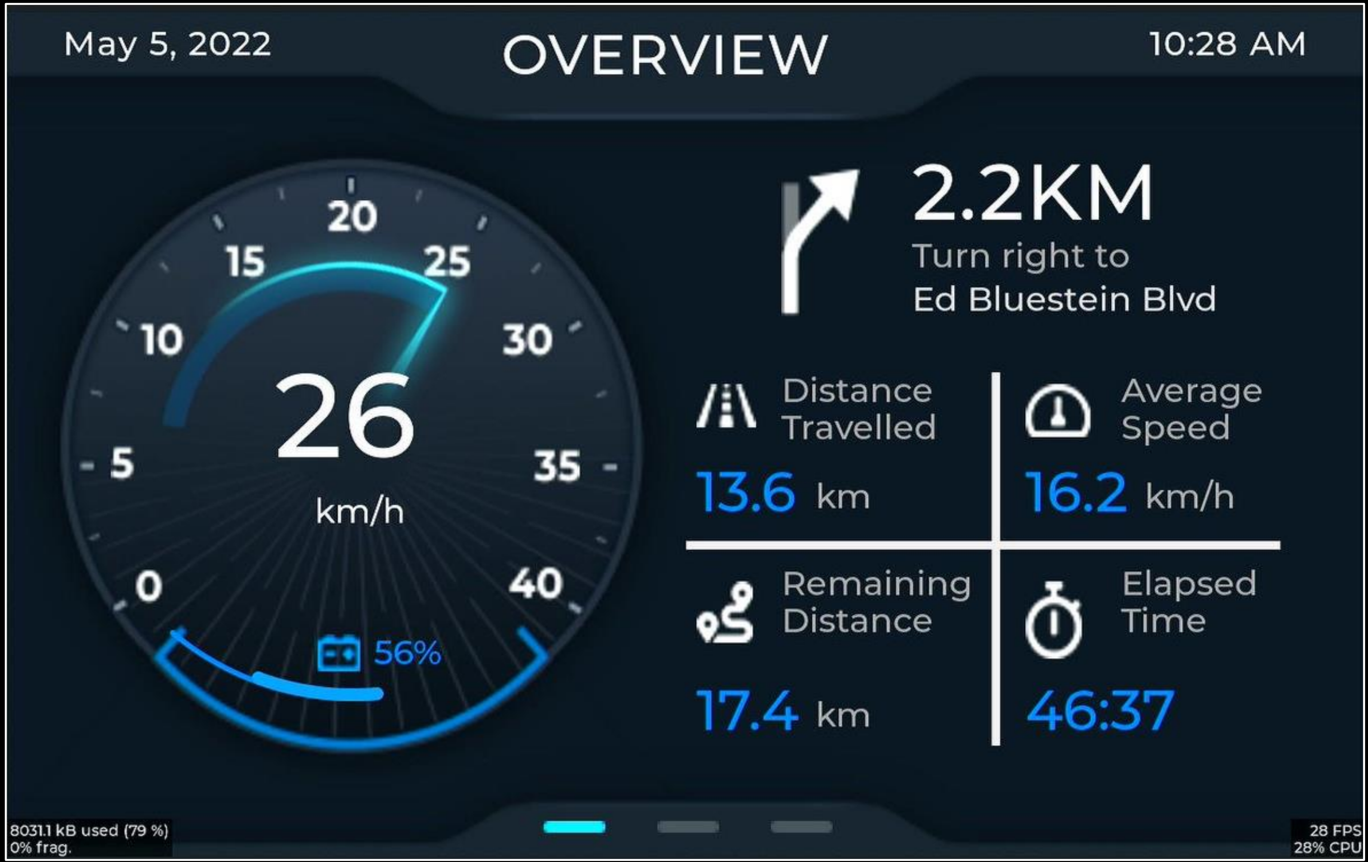
i.MX E-Bike VIT Demo

- i.MX E-Bike VIT showcases the Multimedia capabilities of i.MX to emulate an interactive ebike through a GUI controlled by voice commands. The GUI is based on LVGL (Little Versatile Graphic Library) and NXP's Voice Intelligent Technology (VIT) supports the voice commands.
- Usage: Keyword + command

Wake words supported
HEY NXP
HEY E Bike

Voice Commands supported

NEXT PAGE
LAST PAGE
RUN DEMO
STOP DEMO



Useful Link

- GoPoint User Guide: https://www.nxp.com/webapp/sps/download/preDownload.jsp?render=true
- GoPoint repo: https://github.com/nxp-imx-support/nxp-demo-experience-demos-list/tree/lf-6.12.49-2.2.0 (Including source code of demo: Selfie Segmenter, DMS, ML benchmark, Video test)
- Image Classification/Object Detection: https://github.com/nxp-imx/eiq-example/tree/lf-6.12.49-2.2.0
- i.MX Smart Fitness: https://github.com/nxp-imx-support/imx-smart-fitness
- i.MX Smart Kitchen: https://github.com/nxp-imx-support/smart-kitchen
- i.MX E-Bike VIT: https://github.com/nxp-imx-support/imx-ebike-vit

Model Zoo On i.MX93

Model	Task	Dataset
Deepface-emotion	Image classification	FER2013
Facenet512	Face recognition	LFW
Ultraface-Slim	Object detection	Widerface
SSDLite MobileNetV2	Object detection	COCO
MobileNet V1	Image classification	ImageNet
MoveNet	Pose estimation	COCO + Active
YOLOv4-tiny	Object detection	COCO
Selfie-Segmenter	Semantic Segmentation	Proprietary
WHENet	Pose estimation (head)	300W-LP
facial-landmarks-35-adas-0002	Pose estimation (face)	Proprietary
Tiny-ResNet	Image classification	CIFAR-10
MobileNet V2	Image classification	Imagenet
EfficientNet-lite	Image classification	Imagenet
Visual Wake Word	Image classification	VW COCO2014
EegTCNet	Others / classification	4 motor imagery
MicroSpeech LSTM	Command recognition	Speech commands
Keyword spotting	Command recognition	Speech commands
Anomaly detection AD	Anomaly detection	ToyADMOS
FaceDet	Object detection	WiderFace
MiDaS v2.1 Small	Mono Depth Estimation	10 datasets
wav2letter	Speech recognition	LibriSpeech
ResNet50	Image classification	Imagenet
Fast-SRGAN	Super resolution	DIV2k
YOLACT-Edge	Instance segmentation	COCO
SCI	Low-light enhancement	LOL
CenterNet	Object detection	COCO
Inception v4	Image classification	Imagenet
Yolov8	Object detection	COCO
Yolov5	Object detection	COCO
EfficientDet-lite0	Object detection	COCO

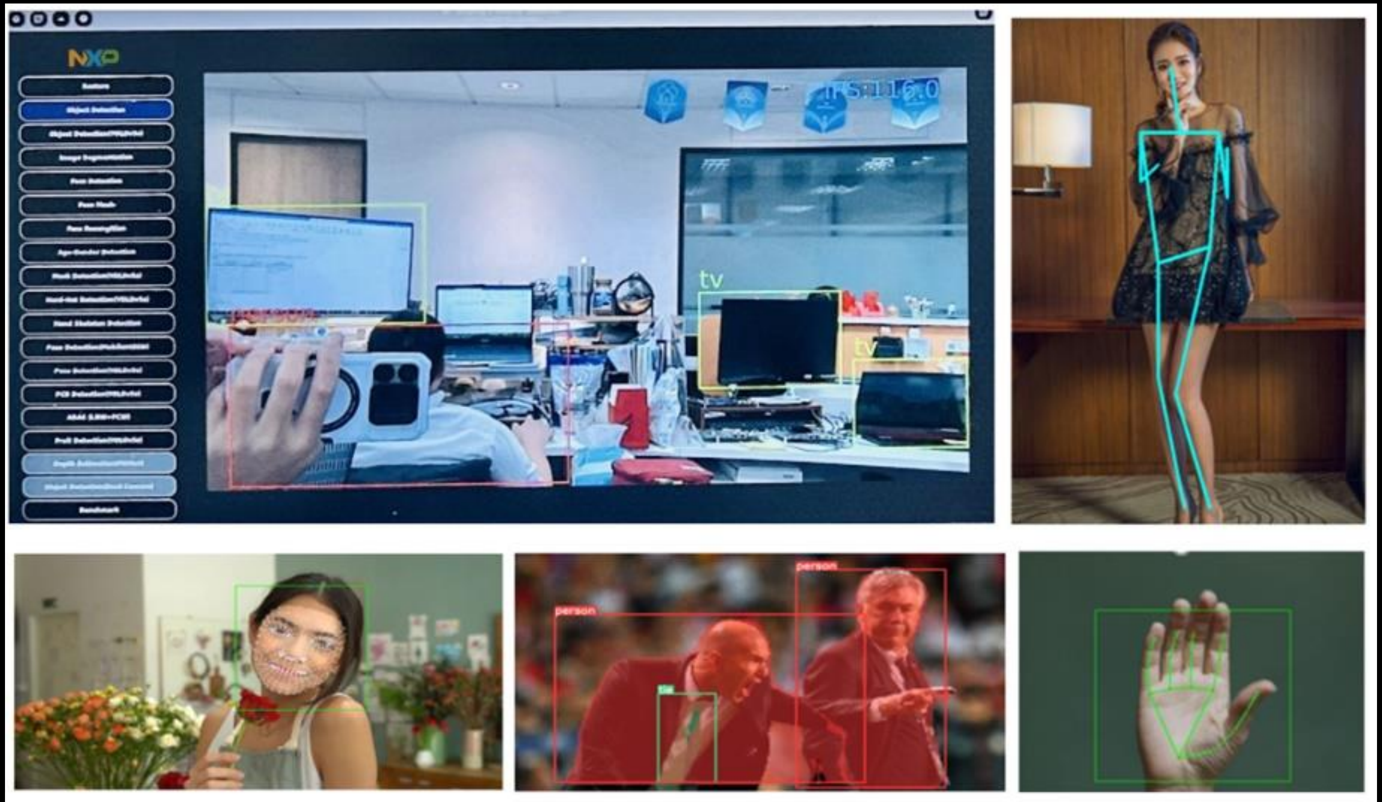
https://github.com/NXP/eiq-model-zoo/tree/main/products

Partner’s Demo on i.MX93

AI Camera

- On-chip NPU (Arm Ethos-U65) for accelerate AI.
- On-chip 2D GPU for image processing.
- Support for a wide variety of peripheral interfaces

		Inference time	
Model Name	Input Size	NPU	CPU
ResNet18	224x224	99.49 ms	249.34 ms
MobilenetV3	224x224	1.99 ms	24.60 ms
MobilenetV2-SSD	300x300	9.95 ms	80.46 ms
YOLOv4-Tiny	416x416	26.20 ms	406.24 ms
YOLOv5s	256x256	20.53 ms	141.73 ms
YOLOv5s-seg	256x256	30.79 ms	198.00 ms



Partner's model zoo

- https://github.com/weilly0912/ATU_NXP_AI_DEMO/tree/v11.0/TensorflowLite_DEMO
- https://github.com/weilly0912/ATU_Model_Zoo/tree/main

Network Name	mAP	Quantized	Input Resolution (HxWxC)	Params (M)	OPS (G)	Pretrained	Quant	Quant(Vela)	Source
yolact_regnetx_1.6gf	27.57	27.27	512x512x3	30.09	125.34	download	download	download	link
yolact_regnetx_800mf	25.61	25.5	512x512x3	28.3	116.75	download	download	download	link
yolov5l_seg	39.78	39.09	640x640x3	47.89	147.88	download	download	download	link
yolov5m_seg	37.05	36.32	640x640x3	32.60	70.94	download	download	download	link
yolov5n_seg	23.35	22.24	640x640x3	1.99	7.1	download	download	download	link
yolov5s_seg	31.57	30.49	640x640x3	7.61	26.42	download	download	download	link
yolov8m_seg	40.6	39.88	640x640x3	27.3	104.6	download	download	download	link
yolov8n_seg	30.32	29.68	640x640x3	3.4	12.04	download	download	download	link
yolov8s_seg	36.63	35.8	640x640x3	11.8	40.2	download	download	download	link

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